



(L-R) Duke Banerjee and Sumit Goswami, Founders, OneArvo Ventures Private Limited

that serves as a unique identity for every product. These patterns can be printed on packaging, labels, cartons, stickers, or bottles, or embedded directly on metallic products via laser marking.

“Although the pattern appears ordinary to the naked eye, it incorporates complex visual elements such as controlled noise, micro-level distortions, and layered variations that are extremely difficult to replicate,” says Banerjee. “Any attempt to copy the pattern through scanning, photography, or reprinting results in subtle information loss, which is not visible to humans but can be accurately detected by ARVO’s computer vision and machine learning algorithms using a smartphone camera,” he adds.

According to Banerjee, each pattern acts as a digital product passport uniquely associated with an individual product. In addition to verifying authenticity, it links and stores key information throughout the product’s life cycle, including manufacturing details, batch information, certifications, logistics tracking, and distribution records.

As products move from factories to distributors, retailers, and ultimately end users, stakeholders can instantly access relevant information

by scanning the pattern, thereby reducing paperwork and manual intervention.

ARVO works with widely used printing technologies, including thermal inkjet, continuous inkjet, fibre laser, handheld printers, and standard label printers. Brands can also integrate ARVO’s verification capability into their own mobile applications, enabling an easy-to-adopt white-label system.

The company has built an in-house lab in Noida equipped with a wide range of printers, inks, substrates, labels, packaging materials, and metallic alloys. This facility enables simulation of real-world manufacturing environments, generation of diverse datasets, and model training across different materials, printers, and environmental conditions.

ARVO currently operates with a team of around 15 full-time employees, including data scientists, engineers, and sales and marketing professionals, with most development and testing handled internally.

The company has deployed over 300,000 codes into the market and has begun generating revenue in the current financial year.

Banerjee explains that the company operates on a

business-to-business, recurring, volume-based revenue model, under which clients pay for platform access along with an annual fee linked to code volume.

ARVO has secured two commercial clients and is running multiple paid proofs of concept with large, tier-one brands, primarily in the automotive sector. It has also completed a significant pilot with Jan Aushadhi medicines under the Pradhan Mantri Bharatiya Janaushadhi Pariyojana.

The company is currently serving clients in the automotive sector and is expanding into the pharmaceuticals, chemicals, fertilisers, electronics, and EMS sectors. While the solution is sector-agnostic by design, ARVO is focusing on select industries at a time to better understand regulatory and operational nuances.

ARVO has filed a patent and is in the process of filing additional patents covering multiple product variants, including tamper-evident labels, metallic authentication, and patterns that remain verifiable even when partially damaged.

Explaining the origin of the company’s name, Banerjee notes that ‘ARVO’ derives from a Finnish word associated with clarity and transparency, symbolising the clearest time of day. The brand positioning centres on authenticity and trust, signalling verified product identity.

Having recently closed a pre-seed funding round, ARVO Ventures plans to strengthen its technology team in the short term and expand its sales and marketing efforts over the next two years. The company is also exploring partnerships with printer manufacturers, packaging vendors, and channel partners to accelerate market adoption.

With plans to enter international markets such as the EU and the US, ARVO is positioning itself as a scalable, cost-effective anti-counterfeiting solution for global supply chains. **EFY**

—Nitisha Dubey